

FINAL HC-POD INTEGRATION WALKTHROUGH

HealthConnect: From Appointment Data to Integrated Decision Support

1. Original Business Problem

HealthConnect is addressing the problem of **missed healthcare appointments and inefficient appointment utilisation**.

Within the project dataset of **5,000 appointments**, 2,423 appointments were recorded as no-shows, giving a validated no-show rate of **48.46%**.

The HC-POD therefore aimed to develop an integrated solution that could:

- understand appointment and no-show patterns;
- identify factors associated with missed appointments;
- investigate whether those factors could support prediction;
- organise the predictive process into a usable technical workflow;
- provide an accessible administrative AI interface; and
- support evidence-based operational decision-making.

2. Different Track Contributions

The final HealthConnect solution combines five connected contributions.

Track	Main contribution
Data Analytics	Data cleaning, validation, exploratory analysis, KPIs, dashboard and business insights
Data Science	Predictive modelling and identification/evaluation of potentially important predictive features
ML Engineering	Structured the model/process into a technical workflow suitable for repeatable use
Generative AI	Administrative conversational support and user interaction
Project Management	Coordination, dependencies, deliverables, integration and final presentation

Each track addresses a different stage of the solution.

3. How the Components Connect

The final workflow is:

Appointment Data



Data Analytics

Clean → Analyse → Validate → Identify patterns



Data Science

Use relevant features → Train/evaluate model → Produce predictive output



ML Engineering

Organise data/model/prediction process into a repeatable technical workflow



Generative AI

Provide an accessible administrative interaction layer



HealthConnect Decision Support

Support healthcare administrative investigation and decision-making



Project Management

Coordinates the workstreams, dependencies, testing, refinement and delivery across the entire process.

Project Management is therefore not simply the final step; it provides coordination across the workflow.

4. How Information and Outputs Moved Between Tracks

The HC-POD demonstrated integration through exchanges of actual outputs.

Data Analytics → Data Science

Data Analytics provided:

- validated appointment KPIs;

- no-show findings;
- previous no-show analysis;
- booking lead-time analysis;
- distance analysis;
- reminder-channel analysis;
- appointment-type findings;
- variable definitions;
- analytical limitations.

These outputs helped provide an evidence base for the predictive modelling work.

Data Science → Data Analytics

Data Science provided predictive modelling findings.

Data Analytics then compared the predictive output with the descriptive findings to understand whether the model supported, differed from or added to the patterns identified in the historical data.

Data Science → ML Engineering

The predictive model/process provides the technical artefact that ML Engineering can organise into a repeatable workflow.

ML Engineering → GenAI / Solution Layer

The technical workflow provides the foundation through which model-related outputs can potentially be made available to downstream applications.

Project Management → All Tracks

Project Management coordinated:

- dependencies;
- deadlines;
- deliverables;
- communication;
- integration activities;
- evidence collection; and
- final presentation preparation.

This demonstrates that the project was developed as a connected HC-POD rather than as five independent assignments.

5. Testing and Refinement

Testing and refinement were important because the first analytical output was not simply accepted without validation.

During the Data Analytics validation process, the KPI displayed as **No-Show Rate = 5.26%** was identified as inconsistent with the underlying data.

The calculation was checked:

No-shows = 2,423

Total appointments = 5,000

Therefore:

$2,423 \div 5,000 \times 100 = 48.46\%$

The **5.26%** figure actually represented:

$263 \text{ cancellations} \div 5,000 \times 100 = 5.26\%$

The dashboard was therefore refined so that:

- **No-Show Rate = 48.46%**
- **Cancellation Rate = 5.26%**

This demonstrates an actual **test → identify issue → validate calculation → refine output** cycle.

The same principle applies to the wider HC-POD: analytical findings can be checked against predictive results, technical workflows can be tested, and the final solution can be refined before presentation.

6. Key Results

The final validated Data Analytics results include:

Measure	Result
Total appointments	5,000
No-shows	2,423
No-show rate	48.46%
Attended	2,314

Measure	Result
Attendance rate	46.28%
Cancellations	263
Cancellation rate	5.26%

Important observed patterns included:

Previous no-show

- Previous no-show: **55.41%**
- No previous no-show: **43.51%**

Booking lead time

- 0–7 days: **27.81%**
- 8–14 days: **33.55%**
- 15–30 days: **43.21%**
- 31–60 days: **60.49%**

Distance

The above-30-km group recorded an observed no-show rate of approximately **68.06%**.

These findings provide candidate areas for predictive investigation and operational review.

7. Final Proposed HealthConnect Workflow

The integrated HealthConnect workflow can be presented as:

STEP 1 — Data

Appointment information enters the HealthConnect analytical process.



STEP 2 — Data Analytics

The data is cleaned, explored and validated.

The analytics layer answers:

What is happening?



STEP 3 — Data Science

Relevant variables are assessed through predictive modelling.

The predictive layer asks:

Can available information help identify potential no-show risk?



STEP 4 — ML Engineering

The model and processing steps are organised into a structured technical workflow.

The engineering layer answers:

How can this process be implemented consistently?



STEP 5 — Generative AI

An administrative AI interface can help users interact with relevant information and administrative processes.

The GenAI layer answers:

How can users interact with the system more naturally?



STEP 6 — Decision Support

The combined outputs provide evidence that can support healthcare administrative planning and investigation.



STEP 7 — Monitoring and Refinement

Results, model performance, data quality and user feedback can be reviewed to improve the solution over time.

8. Business Value

The integrated solution could provide HealthConnect with several potential benefits.

Improved visibility

The dashboard provides a clearer view of appointment outcomes and important patterns.

Risk identification

Predictive modelling could help identify appointment characteristics associated with higher observed no-show risk.

Better operational investigation

HealthConnect can investigate whether scheduling, reminder and follow-up processes should be adapted for particular appointment segments.

Better use of data

Historical appointment information is transformed into structured evidence for decision-making.

Improved administrative experience

Generative AI can provide a more accessible way for users to interact with administrative information.

Integrated decision support

The major benefit of the HC-POD approach is that analytics, prediction, engineering and AI are connected into one workflow rather than operating independently.

9. Limitations

The current HealthConnect solution has important limitations.

Data limitations

The analysis is based on the available project dataset and may not capture every factor affecting appointment attendance.

Causality

The observed analytical relationships do not prove that any particular factor causes a patient to miss an appointment.

Prediction limitations

A machine-learning prediction represents an estimated risk and cannot guarantee an individual patient's future behaviour.

AI limitations

Generative AI can produce incorrect or inappropriate responses and therefore requires suitable controls, validation and human oversight.

Deployment limitations

The project represents a proposed solution. A real-world deployment would require additional testing, infrastructure, security, privacy controls and operational validation.

Healthcare limitations

The solution is intended to support administrative decision-making and should not replace professional clinical judgement.

10. Recommendations

Based on the integrated work, the HC-POD recommends:

1. **Continue validating the predictive model** using appropriate evaluation metrics and testing.
2. **Investigate previous no-show behaviour, booking lead time and distance** as potentially useful risk indicators.
3. **Evaluate reminder strategies more rigorously** before making causal claims about channel effectiveness.
4. **Maintain clear KPI definitions and validation procedures** to prevent dashboard interpretation errors.
5. **Develop appropriate privacy, security and responsible-AI controls** before any real-world deployment.
6. **Monitor model and system performance continuously** if the solution progresses toward implementation.
7. **Conduct further real-world validation** before using predictions to influence operational decisions.

11. Final HC-POD Integration Statement

The HealthConnect HC-POD demonstrates an end-to-end multidisciplinary workflow.

Data Analytics transforms appointment data into validated evidence and identifies important patterns.

Data Science builds on those findings through predictive modelling.

ML Engineering organises the predictive process into a structured technical workflow.

Generative AI provides an administrative interaction layer.

Project Management coordinates the dependencies, exchanges, testing, refinement and delivery across the workstreams.

The result is a connected HealthConnect concept that moves from:

Data → Analytics → Prediction → Engineering → AI Interaction → Decision Support

The integration is demonstrated not only by the presence of five different track outputs, but by the **exchange, testing, refinement and use of those outputs across the final solution**.